

Electricity Price Formation & Merit Order Analysis

Germany • 2015–2024 • 10 Years • 29,217 Observations

Tools: Python • pandas • matplotlib • numpy • Jupyter Notebook

Dataset: [European Power Markets 2015–2024 — Synthetic Merit Order Dataset \(Kaggle\)](#)

Key Metrics at a Glance

Metric	Value	What It Means
Total Observations (Germany)	29,217	3-hourly snapshots across 10 years — 8 readings per day
Gas Crisis Multiplier	26.6×	2022 annual median (310.9 EUR/MWh) vs 2019–2020 pre-crisis median (11.7 EUR/MWh)
Peak Gas Marginal Cost (2022 avg)	552 EUR/MWh	Up from 46 EUR/MWh in 2019 — merit order amplified this 12× gas rise into a 26.6× electricity price rise
Wind Scenario Price Impact	−10.2 EUR/MWh	Average reduction in monthly median prices under the +50% wind scenario
Negative Price Hours 2015–2024	528 hours	Peaked at 144 in 2018; fell to exactly zero in 2022 and 2023
Post-Crisis Price Floor (2024)	38.6 EUR/MWh	3.3× above pre-crisis — carbon price (57 EUR/tCO ₂) now structurally elevated

1. Project Overview

This project analyses how wholesale electricity prices are determined in the German power market using 10 years of synthetic European 3-hourly electricity market data (2015–2024). The analysis focuses on the Merit Order mechanism - the pricing framework through which power plants are dispatched from lowest to highest marginal cost, with the final plant required to meet demand setting the market clearing price.

The study covers 2015–2024, with particular focus on the 2022 European gas crisis - a period of extreme volatility following Russia's invasion of Ukraine. During this period, wholesale electricity prices rose 26.6× above pre-crisis levels as surging gas and carbon prices significantly increased the marginal cost of gas-fired generation, reshaping power market economics across Europe.

Central Question

How are wholesale electricity prices determined in a liberalised power market, and how did the 2022 gas supply crisis transmit into electricity prices through the merit order mechanism?

Why It Matters

- The 2022 energy crisis cost European governments over €700 billion in subsidies. This project explains mechanistically why the crisis was so severe.
- Findings directly inform current policy debates on battery storage investment, renewable expansion, and energy security.

2. Dataset Summary

- The dataset is a synthetic European-style electricity market simulation covering 2015–2024, published on Kaggle. It was built on a merit order model with realistic commodity price linkages and includes deliberately encoded real-world events: the 2022 European gas crisis (gas prices simulated to a peak of €350/MWh), the COVID-2020 demand collapse (~20% reduction), and renewable surplus negative price events.
- While synthetic, the dataset accurately replicates the microeconomic mechanics of liberalised power markets, making it suitable for merit order analysis and price dynamic studies.
- Three of the five available files were used. All three contained zero missing values across all columns.

Dataset	Coverage	Purpose
generation_mix.csv	146,082 total rows (5 countries) 29,217 after filtering to Germany 3-hourly	Generation by fuel type (MW) - nuclear, hydro, wind, solar, gas, coal per timestep. Supply side of the electricity market.
power_prices.csv	146,085 total rows 29,217 (DE) 3-hourly	Day-ahead wholesale price (EUR/MWh), total demand, capacity factors, negative price flag. Demand side and market outcome.
commodity_prices.csv	3,653 rows daily	Gas, coal, and carbon permit prices - inputs to marginal cost calculations. Also contains pre-calculated spread columns (spark_spread, dark_spread, clean_spark_spread) and a shock_event flag labelling crisis periods.

Note: The dataset also includes demand_features.csv (daily demand with temperature/HDD features) and price_zones.csv (zone metadata for all 5 countries), which were not used in this analysis.

3. Key Terms

Term	Definition
Merit Order	The system by which power plants are ranked cheapest to most expensive and dispatched in that order. The most expensive plant still needed to meet demand sets the market price for all generators that hour.
Marginal Cost (EUR/MWh)	The cost of producing one additional MWh from a given plant, including fuel and carbon costs but not capital costs. Wind and solar have zero marginal cost.
Marginal Plant	The most expensive power plant still running to meet demand at a given hour - this plant sets the market clearing price. In Germany, gas frequently acts as the marginal plant during periods of higher demand.
Market Clearing Price	The single price at which supply equals demand in a given hour. All generators - including cheap wind farms - receive this price regardless of their own production cost.
Merit Order Effect	The phenomenon where increasing renewable generation displaces higher-cost plants (gas), reducing the market clearing price. More wind = less gas on the margin = lower electricity price.
Negative Price	When the electricity price falls below zero EUR/MWh. Occurs when wind/solar generate more than the grid can consume — generators pay consumers to take electricity rather than shut down.
Spark Spread (EUR/MWh)	Electricity price minus gas fuel cost. A positive spread means it is profitable to run a gas plant; negative means it is losing money per MWh produced.
Supply Stack	A chart showing all available generation capacity arranged in merit order - cheapest on the left, most expensive on the right. Where the demand line intersects identifies the marginal plant and clearing price.

4. Exploratory Data Analysis (Python)

Before any analysis, all three datasets were loaded and examined to understand their structure, date ranges, column types, and data quality. The EDA confirmed that the datasets were clean, complete, and correctly structured for merging.

4.1 Libraries & Data Loading

Code

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

gen      = pd.read_csv('generation_mix.csv',    parse_dates=['datetime'])
prices   = pd.read_csv('power_prices.csv',      parse_dates=['datetime'])
comm     = pd.read_csv('commodity_prices.csv',  parse_dates=['date'])
```

Output

```
generation_mix shape : (146082, 11)
power_prices shape   : (146085, 11)
commodity_prices shape: (3653, 9)
```

4.2 Structural Exploration - generation_mix

The generation_mix dataset was inspected with .head(), .info(), and .isnull().sum() to confirm structure and data quality before filtering.

Code

```
print(gen.head())
gen.info()
print('Zones:', gen['zone'].unique())
print('Date range:', gen['datetime'].min(), 'to', gen['datetime'].max())
print('Missing values:', gen.isnull().sum())
```

Check	Finding
Zones	5 countries - DE, FR, UK, ES, IT. Analysis scoped to Germany (DE) only.
Date range	2015-01-01 to 2024-12-31 - full 10-year window, no gaps
Interval	3-hourly (00:00, 03:00 ... 21:00) = 8 readings per day per country
Missing values	Zero missing values across all 11 columns
Units	All generation columns in MW (nuclear_mw, wind_mw, solar_mw, gas_mw, coal_mw)
Derived column	renewables_pct = (wind + solar + hydro) / total × 100 - pre-calculated in dataset

4.3 Power Prices Exploration

The `power_prices` dataset was inspected with `.head()`, `.info()`, and `.isnull().sum()` to confirm structure and data quality before filtering. This dataset holds the primary dependent variable - the wholesale electricity price and the demand and negative price columns used throughout the analysis.

Code

```
print(prices.head())
prices.info()
print('Zones:', prices['zone'].unique())
print('Date range:', prices['datetime'].min(), 'to', prices['datetime'].max())
print('Missing values:', prices.isnull().sum())
print('Price stats:', prices['price_eur_mwh'].describe().round(2))
```

Check	Finding
Zones	5 countries - DE, FR, UK, ES, IT. Analysis scoped to Germany (DE) only.
Date range	2015-01-01 to 2024-12-31 - full 10-year window, no gaps
Interval	3-hourly (00:00, 03:00 ... 21:00) = 8 readings per day per country
Missing values	<code>shock_event</code> has 108,325 NaN — expected, NaN = normal market conditions. All analytical columns 100% complete
Row count match	146,085 total; 29,217 after DE filter - exact match with <code>generation_mix</code> confirming clean timestamp alignment
Price distribution	Right-skewed: mean (65.60) >> median (22.52 EUR/MWh) - 2022 crisis prices pull mean significantly above median
Max price	2,601.80 EUR/MWh - extreme scarcity spike encoded in the dataset
Columns used	<code>price_eur_mwh</code> (primary dependent variable), <code>demand_mw</code> (supply stack), <code>is_negative_price</code> (negative price counting)
Columns not used	<code>wind_cf</code> , <code>solar_cf</code> , <code>net_demand_frac</code> - generation volumes taken from <code>generation_mix</code> instead

4.4 Commodity Price Exploration

The `commodity_prices` dataset holds the daily input costs that drive the merit order calculation. Its explored to understand price ranges and identify the scale of the 2022 gas crisis. The dataset also contains pre-calculated spread columns (`spark_spread`, `dark_spread`, `clean_spark_spread`), an `oil_price_usd_bbl` column (not used - no oil-fired generation in Germany's merit order), and a `shock_event` flag labelling major market disruptions.

Code

```
comm['year'] = comm['date'].dt.year
yearly = comm.groupby('year')[['gas_price_eur_mwh',
                              'coal_price_usd_t',
                              'carbon_price_eur_t']].mean().round(1)

print(yearly)
```

Output

Yearly average gas price			
	gas_price_eur_mwh	coal_price_usd_t	carbon_price_eur_t
year			
2015	14.8	68.6	25.3
2016	10.2	53.7	12.8
2017	10.8	66.3	11.7
2018	12.3	67.5	7.5
2019	21.9	77.4	11.1
2020	15.7	70.9	10.7
2021	18.3	81.8	11.3
2022	269.6	59.9	66.0
2023	105.6	39.7	55.5
2024	46.3	55.1	57.2

Check	Finding
Rows	3,653 - one row per calendar day, 2015-01-01 to 2024-12-31
Columns	9 - date, gas, coal, carbon, oil, spark_spread, dark_spread, clean_spark_spread, shock_event
Missing values	shock_event has NaN for normal market periods - all price columns 100% complete
Gas price range	5.7 to 350.0 EUR/MWh - peaked September 2022 (Ukraine war energy crisis)
Carbon price range	6.4 to 80.9 EUR/tCO ₂ - rising trend as EU tightened ETS carbon supply
Columns used	gas_price_eur_mwh, coal_price_usd_t, carbon_price_eur_t
Columns not used	oil_price_usd_bbl, spark_spread, dark_spread, clean_spark_spread - not required for marginal cost calculations

This output confirmed the central analytical observation: gas rose from 21.9 EUR/MWh (2019) to 269.6 **EUR/MWh** (2022) - a 12× increase. Carbon simultaneously rose ~6×. Gas peaked at 350 EUR/MWh in September 2022

5. Data Preparation & Feature Engineering

Three preparation steps converted the raw datasets into a single analysis-ready master table: country filtering, dataset merging, and marginal cost calculation.

5.1 Country Filter - Germany Only

The raw datasets span 5 European zones (DE, FR, UK, ES, IT). Germany was selected as the focus market due to its size, fuel diversity, and prominence during the 2022 gas crisis. Both `generation_mix` and `power_prices` were filtered to `zone = 'DE'`, reducing from ~146,000 rows to 29,217.

Code

```
gen_de = gen[gen['zone'] == 'DE'].copy()
prices_de = prices[prices['zone'] == 'DE'].copy()

print('Germany generation rows:', len(gen_de))
print('Germany price rows      :', len(prices_de))
```

Output

```
Germany generation rows: 29217
Germany price rows      : 29217
```

5.2 Merging the Three Datasets

Two sequential merges were performed to build the master DataFrame:

- Join 1 - `generation_mix` (DE) and `power_prices` (DE) were merged on the datetime column using an inner join. Both datasets share identical 3-hourly timestamps, ensuring a clean 1:1 match across all 29,217 rows.
- Join 2 - `commodity_prices` was merged onto the master table using a left join on date. Since commodity prices are recorded daily, each day's single price value is applied across all 8 intraday readings. A left join was used to ensure no generation or price rows were dropped in the event of a missing commodity date.

Code

```
df = gen_de.merge(prices_de[['datetime', 'price_eur_mwh',
                             'demand_mw', 'is_negative_price']],
                  on='datetime', how='inner')

df['date'] = df['datetime'].dt.date
comm['date_key'] = comm['date'].dt.date

df = df.merge(comm[['date_key', 'gas_price_eur_mwh',
                   'coal_price_usd_t', 'carbon_price_eur_t']],
              left_on='date', right_on='date_key', how='left')

df['year'] = df['datetime'].dt.year
df['month'] = df['datetime'].dt.month
df['hour'] = df['datetime'].dt.hour

print(df.shape)  # → (29217, 24)
```

5.3 Marginal Cost Calculation (Feature Engineering)

- The two most important derived columns - `gas_mc` and `coal_mc` , are the marginal cost of producing 1 MWh of electricity from gas and coal respectively
- These columns are not present in the raw data - they are derived from daily commodity prices using standard power market engineering parameters. The formula for each fuel accounts for two costs: the fuel input cost (adjusted for plant thermal efficiency) and the carbon permit cost under the EU Emissions Trading System (ETS).

Fuel	Parameter	Value	Rationale
Gas	Thermal efficiency	÷ 0.50	A gas CCGT plant converts ~50% of fuel energy into electricity; remaining 50% is waste heat
Gas	Emission factor	0.20 tCO ₂ /MWh	Gas emits ~0.20t CO ₂ per MWh generated multiplied by the carbon price (EU ETS)
Coal	Energy content	÷ 8.0 MWh/t	1 tonne of thermal coal yields ~8 MWh of heat energy
Coal	Currency	× 0.92 USD→EUR	Coal is priced in USD/tonne; converted to EUR
Coal	Emission factor	0.85 tCO ₂ /MWh	Coal is ~4× dirtier than gas — high carbon prices hit coal economics severely
Wind / Solar	Marginal cost	0 EUR/MWh	Fuel is free - completely insulated from commodity price shocks
Nuclear / Hydro	Marginal cost	5 / 2 EUR/MWh	Near-zero operational fuel cost; small values reflect dispatch priority

Code

```
USD_TO_EUR = 0.92

def calc_marginal_costs(gas_eur_mwh, coal_usd_t, carbon_eur_t):
    coal_mc = (coal_usd_t / 8.0) * USD_TO_EUR + 0.85 * carbon_eur_t
    gas_mc = gas_eur_mwh / 0.50 + 0.20 * carbon_eur_t
    return {
        'wind': 0.0, 'solar': 0.0, 'hydro': 2.0, 'nuclear': 5.0,
        'coal': round(coal_mc, 1),
        'gas' : round(gas_mc, 1), 'other' : 40.0, }

for year, label in [(2019, '2019 NORMAL MARKET'), (2022, '2022 GAS CRISIS'), (2024, '2024 RECOVERY')]:
    year_data = comm[comm['year'] == year]
    mc = calc_marginal_costs(year_data['gas_price_eur_mwh'].mean(),
                             year_data['coal_price_usd_t'].mean(),
                             year_data['carbon_price_eur_t'].mean())
    print(f"\n{label}")
    for fuel, cost in mc.items():
        bar = '█' * int(cost / 10)
        print(f" {fuel:<10} {cost:>6.1f} {bar}")
```


Output :

2019 NORMAL MARKET		2022 GAS CRISIS		2024 RECOVERY	
wind	0.0 EUR/MWh	wind	0.0 EUR/MWh	wind	0.0 EUR/MWh
solar	0.0 EUR/MWh	solar	0.0 EUR/MWh	solar	0.0 EUR/MWh
hydro	2.0 EUR/MWh	hydro	2.0 EUR/MWh	hydro	2.0 EUR/MWh
nuclear	5.0 EUR/MWh	nuclear	5.0 EUR/MWh	nuclear	5.0 EUR/MWh
coal	18.3 EUR/MWh	coal	63.0 EUR/MWh	coal	54.9 EUR/MWh
gas	46.1 EUR/MWh	gas	552.4 EUR/MWh	gas	103.9 EUR/MWh
other	40.0 EUR/MWh	other	40.0 EUR/MWh	other	40.0 EUR/MWh

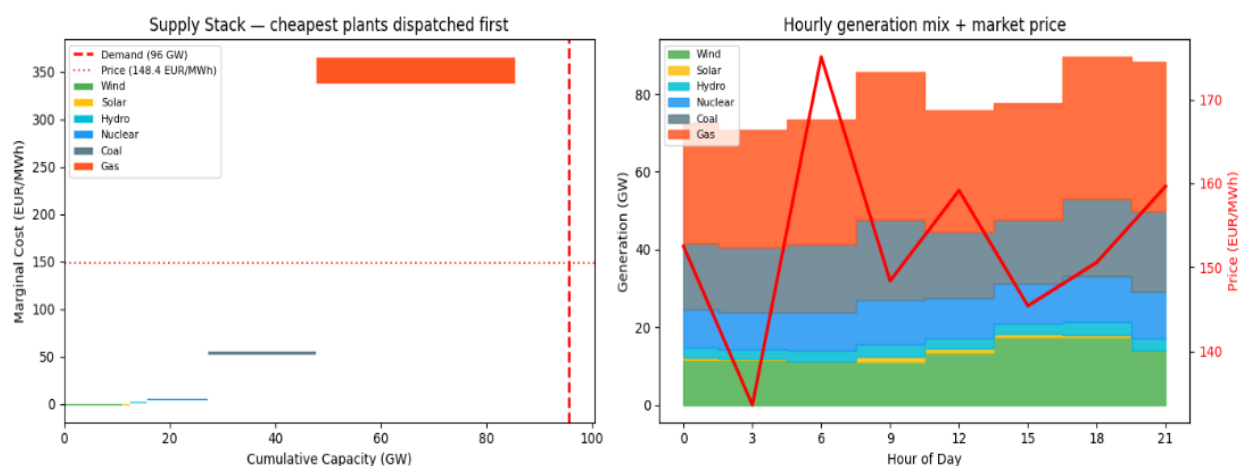
6. Python Analysis Steps

Five structured analytical questions were answered using Python

Q1 What does the merit order supply stack look like for a representative hour?

- A winter weekday morning (January 18, 2023, 09:00) was selected — high heating demand and minimal solar output make gas the unambiguous marginal plant. The supply stack is built by sorting each fuel by marginal cost and plotting cumulative capacity against cost, with the demand line marking where the market clears. At this hour, demand was 95.7 MW, gas marginal cost was 351.7 EUR/MWh, and the market cleared at 148.4 EUR/MWh

Figure 1 — Merit Order Supply Stack | Germany | 2023-01-18 | 09:00



- The gap between market price (148.4 EUR/MWh) and estimated gas marginal cost (351.7 EUR/MWh) suggests that wholesale prices did not fully track fuel costs during this period, likely reflecting government interventions and other market factors active during the energy crisis

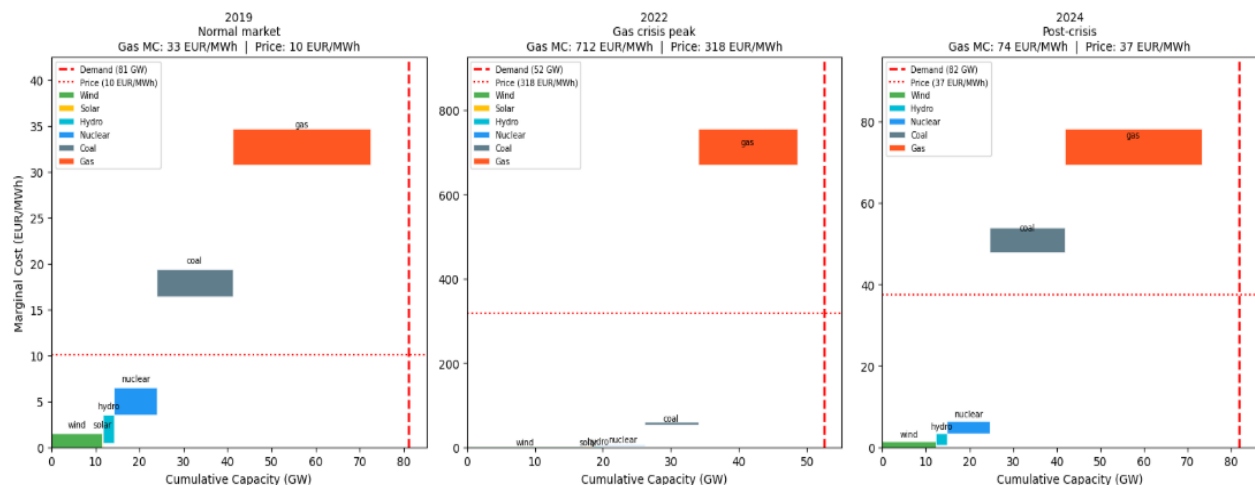
Variable	Value — Jan 18 2023, 09:00
Demand	95.7 MW
Gas marginal cost (that day)	351.7 EUR/MWh
Market clearing price	148.4 EUR/MWh
Price vs gas MC gap	–203.3 EUR/MWh - Observed market price remained below estimated gas marginal cost, likely due to government support measures and market interventions during the energy crisis
Wind output at 09:00	11.1 MW - producing at 0 MC, earns the 148.4 EUR/MWh clearing price
Coal marginal cost (that day)	54.6 EUR/MWh — cheaper than gas but not on the margin

Q2 How did the 2022 gas crisis reshape the supply stack vs normal years?

- Comparing representative winter weekdays across three market regimes demonstrates how commodity price shocks restructure the entire supply stack. Each panel uses an independent y-axis scale (set to $\text{gas_mc} \times 1.3$) so all fuels remain visible even when the 2022 gas bar is 20× taller than 2019.

Note: Gas MC values in the table below reflect the specific days plotted, not yearly averages

Figure 2 — How the Gas Crisis Reshaped the Merit Order | Germany



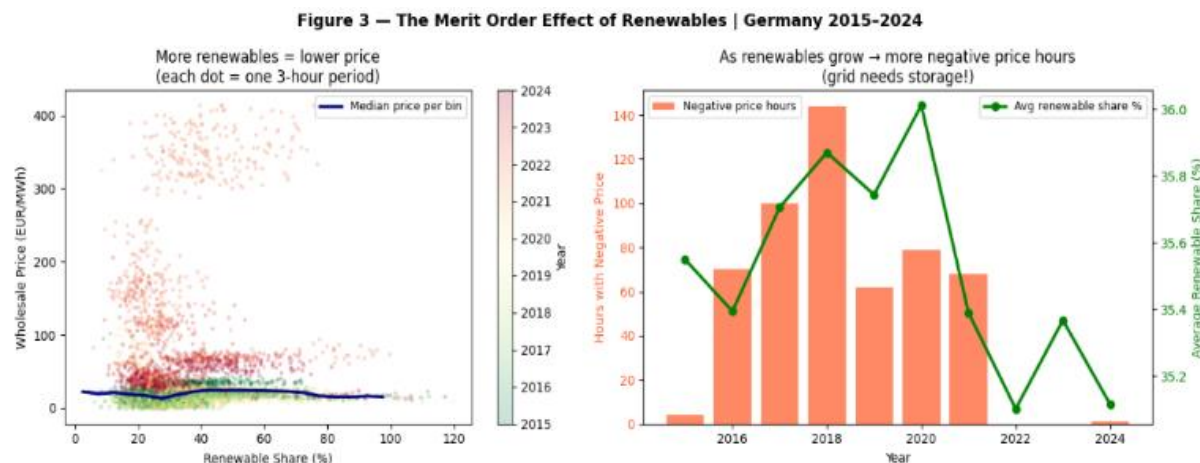
- The 2024 panel confirms the market has structurally reset - the staircase shape has returned but the y-axis scale remains approximately 2× that of 2019, consistent with the post-crisis price floor of 38.6 EUR/MWh observed in the annual data

Day (06:00 snap)	Gas MC	Market Price	Interpretation
2019-01-16	32.7 EUR/MWh	10.0 EUR/MWh	Normal staircase — all fuels visible on the same y-axis scale
2022-09-01	712.0 EUR/MWh	318.4 EUR/MWh	Market price (318) remained substantially below estimated gas MC (712), likely reflecting government interventions and other market factors during the energy crisis. Gas commodity price peaked at 350 EUR/MWh on this date.
2024-01-17	73.7 EUR/MWh	37.3 EUR/MWh	Staircase shape restored but y-axis still ~2× 2019 scale — not fully normalised

Q3 Does higher renewable share reduce the wholesale price?

- The relationship between renewables_pct and price_eur_mwh was explored across all 29,217 observations.

Note: In this synthetic dataset, renewable share is approximately constant at ~35% across all years by design — the chart shows how price changes with renewable share within individual years rather than showing a growth trend.

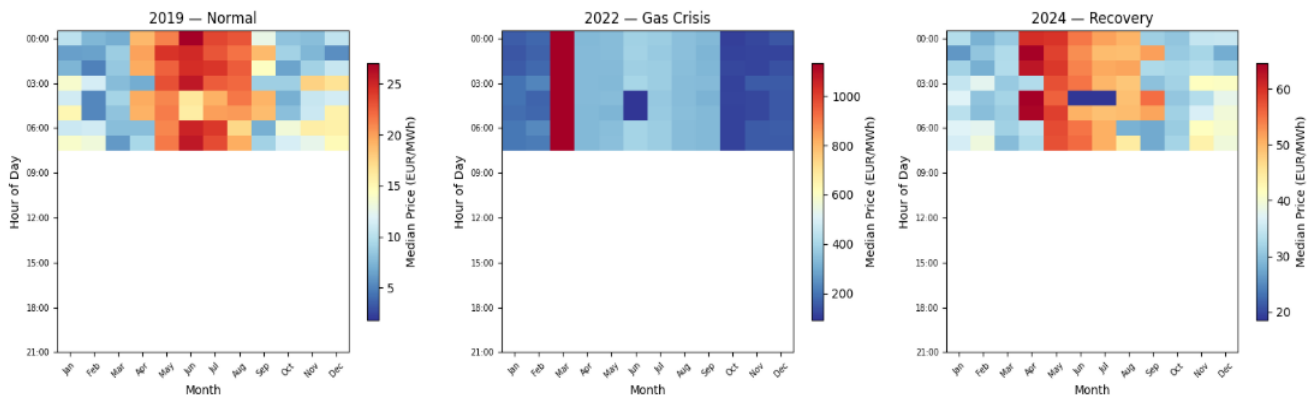


- Within any single year, higher renewable share correlates with lower prices - the merit order effect is confirmed in the data.
- In aggregate, the 2022 data overwhelm this relationship: gas was so expensive that even hours with 60–70% renewable share still had elevated prices.
- Negative price hours peaked at 144 in 2018 and fell to exactly zero in both 2022 and 2023 - crisis-level gas prices made every MWh too valuable to curtail.

Q4 How did price patterns by hour and month change across the three market regimes?

- A heatmap of median price by hour (y-axis, 00:00–21:00) and month (x-axis) was produced for 2019, 2022, and 2024. Each year uses its own colour scale (5th - 95th percentile) so within-year patterns remain visible regardless of absolute price differences.

Figure 4 — Price Heatmap: Hour of Day vs Month | Germany

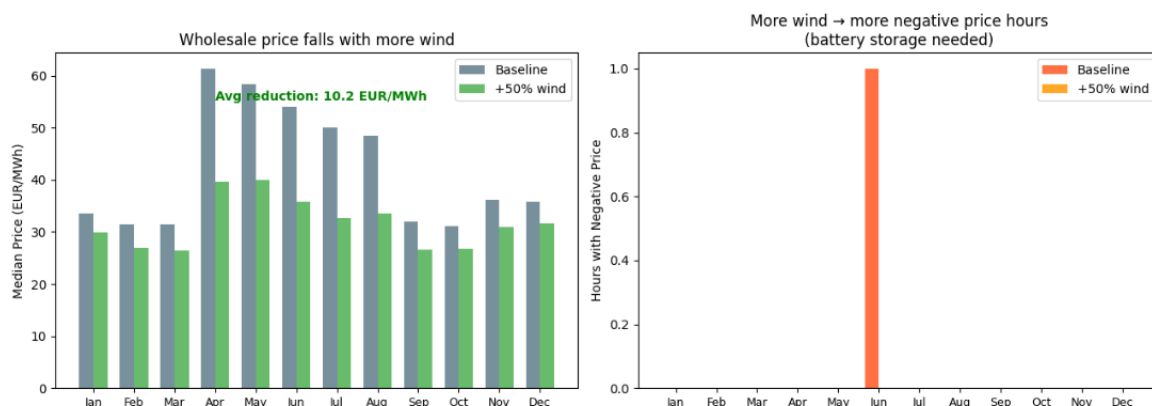


- 2019:** Clear seasonal and daily patterns - expensive winter mornings (06:00–09:00), cheapest summer nights. Price range ~5 - 30 EUR/MWh.
- 2022:** Heatmap almost uniformly deep red across every hour and month. Brief relief in June–July midday from peak solar generation. Price range ~100 - 600 EUR/MWh.
- 2024:** Seasonal and daily patterns restored. Price range settled at ~20–60 EUR/MWh - structurally 2–3× above 2019.

Q5 What would +50% wind capacity do to 2024 wholesale prices?

- Wind generation for 2024 was scaled by 1.5×, and a net demand ratio approach was applied to estimate the resulting market clearing price for each 3-hourly timestep.

Figure 5 — Wind Scenario: +50% Wind Capacity | Germany 2024



- Average price reduction across 2024 monthly medians: 10.2 EUR/MWh - equivalent to ~€5.1 billion in annual consumer savings at Germany's ~500 TWh consumption
- Spring/summer months show the largest reductions (wind is strongest and solar already displaces some gas demand).
- Winter reductions are smaller - demand remains high enough that gas stays on the margin even with 50% more wind.
- The scenario reduced negative price hours from 1 to 0 in the 2024 baseline data. While the model indicates lower wholesale prices under higher wind penetration, managing potential oversupply at higher levels of renewable capacity would likely require additional storage infrastructure

7. Commodity Price History (2015 - 2024)

Yearly average commodity prices confirm the scale of the 2022 crisis. Gas rose 12× from the 2019 baseline (21.9 → 269.6 EUR/MWh) while the EU carbon price simultaneously rose ~6× (11.1 → 66.0 EUR/tCO₂). Both shocks hitting simultaneously is what the merit order amplified into a 26.6× electricity price increase.

Year	Gas (EUR / MWh)	Coal (USD / t)	Carbon (EUR/ t CO ₂)	Market Character
2019	21.9	77.4	11.1	Baseline year for comparisons
2020	15.7	70.9	10.7	COVID demand collapse , ~20% demand reduction
2021	18.3	81.8	11.3	Gas crisis onset -prices beginning to rise
2022	269.6	59.9	66.0	! GAS CRISIS PEAK -- gas hit 350 EUR/MWh in Sep 2022
2023	105.6	39.7	55.5	Partial recovery - prices falling but carbon elevated
2024	46.3	55.1	57.2	New normal - gas recovering but carbon remains structurally higher

8. Results Summary

Finding	Value	Significance
Pre-crisis median electricity price (2019–2020)	11.7 EUR/MWh	Baseline for all multiplier calculations
Crisis peak median price (2022 annual)	310.9 EUR/MWh	26.6× above pre-crisis baseline
Gas MC increase: 2019 → 2022 (annual averages)	46 → 552 EUR/MWh	12× gas rise → 26.6× electricity rise via merit order amplification
Post-crisis price floor (2024 median)	38.6 EUR/MWh	3.3× above pre-crisis - structural shift driven by elevated carbon
Negative price hours 2015–2024	528 total	Peaked 144 hrs in 2018; exactly zero in 2022 and 2023
Carbon: 2019 → 2022	11.1 → 66.0 EUR/tCO ₂	~ 6× increase compounding the gas shock
+50% wind - average price reduction	– 10.2 EUR/MWh	Average reduction in monthly median prices across 2024
Gas MC vs Market Price (Sep 2022)	Market price 318 < Gas MC 712 on Sep 2022	Market prices remained below estimated gas marginal costs during peak crisis conditions.

9. Key Findings

Merit Order Mechanism

- The merit order determined electricity prices consistently across all 10 years. Gas plants frequently acted as the marginal price-setting technology in the German electricity market as seen in Figure 1 supply stack analysis.
- When gas prices rose 12× from the 2019 baseline in 2022, the merit order formula amplified this into a 26.6× electricity price increase - because every generator on the grid (including zero-cost wind) earns the gas-set clearing price.
- The 2022 market clearing price (318 EUR/MWh on the Sep 1 peak day) remained significantly below the estimated gas marginal cost (712 EUR/MWh), suggesting that wholesale prices did not fully track fuel costs during peak crisis conditions, likely reflecting government interventions and other market factors during the energy crisis

Renewable Merit Order Effect

- The merit order effect of renewables - higher renewable share → lower wholesale price - is confirmed in the data within individual years (Figure 3 scatter and binned median analysis).
- In aggregate across all years, 2022 data overwhelms this relationship. Gas was so expensive that even hours with 60–70% renewable share still cleared at elevated prices. This demonstrates that commodity price shocks dominate the renewable merit order effect in crisis conditions.
- Negative price hours fell to exactly zero in 2022 and 2023 - the first time in the 10-year dataset. Crisis-level gas prices made every MWh economically valuable, temporarily eliminating the grid's renewable oversupply problem.

Post-Crisis Structural Shift

- The 2024 median price of 38.6 EUR/MWh is 3.3× the pre-crisis level despite gas partially recovering to 46.3 EUR/MWh. The persistent elevation is driven by the EU carbon price remaining at 57.2 EUR/tCO₂ - compared to 11.1 EUR/tCO₂ in 2019.
- Price heatmaps confirm that daily and seasonal patterns fully restored by 2024, but on a higher absolute scale — winter morning peaks now occur at 40–60 EUR/MWh vs 15–25 EUR/MWh pre-crisis.

Wind Scenario Policy Implication

- +50% wind capacity reduces average 2024 prices by 10.2 EUR/MWh - demonstrating the merit order effect of additional renewables in a post-crisis market where gas still sets the clearing price on most hours.
- While the modelled scenario reduced negative-price hours in the 2024 baseline data, higher levels of renewable penetration may require additional storage capacity to manage future oversupply and maintain system flexibility as renewable generation increases.

10. Recommendations

Energy Security (Policymakers)

Renewable capacity alone does not protect against commodity price shocks in the short run - 35% renewable share in 2022 was insufficient to prevent a 26.6× price spike. The data supports a parallel strategy: accelerating renewable buildout alongside strategic gas storage reserves and demand flexibility mechanisms.

Storage Investment (Grid Operators)

The +50% wind scenario reduced average wholesale prices by 10.2 EUR/MWh while keeping negative-price events limited in the 2024 baseline data. As renewable penetration increases beyond the levels modelled here, additional storage capacity may be required to manage periods of surplus generation and maintain system flexibility.

Renewable Project Economics (Investors)

The post-crisis price floor of ~38.6 EUR/MWh provides significantly stronger project economics for new renewable assets compared to the pre-crisis floor of ~11.7 EUR/MWh. Combined with a carbon price now structurally at ~57 EUR/tCO₂, the merit order consistently favours renewable investors more than the pre-2021 market.

Price Forecasting (Analysts)

Gas price forecasting dominates renewable share analysis for near-term electricity price prediction. Any model excluding commodity price dynamics will systematically underperform - particularly during supply disruption scenarios. The 2022 data shows this asymmetry clearly: gas MC explained far more price variance than renewable share.

11. Conclusion

This project analysed 10 years of German electricity market data (2015–2024) to understand how wholesale electricity prices are influenced by fuel costs, generation mix, and the merit order mechanism.

The results show that gas prices were the primary driver of electricity price volatility during the 2022 energy crisis. A 12× increase in gas prices from the 2019 baseline corresponded with a 26.6× increase in wholesale electricity prices, highlighting how the merit order can transmit commodity price shocks into electricity markets. The analysis also found that while higher renewable generation generally lowers prices through the merit order effect, this impact can be outweighed by extreme commodity price shocks.

The scenario analysis showed that increasing wind capacity by 50% could reduce average wholesale prices by 10.2 EUR/MWh. While the modelled scenario reduced negative-price hours in the 2024 baseline data, higher levels of renewable penetration may require additional storage capacity to manage potential oversupply and maintain system flexibility as renewable generation increases
